

1) What is A , given $A = I \begin{bmatrix} 7 \\ -2 \end{bmatrix}$?

$$\forall \vec{x}, I\vec{x} = \vec{x} \therefore A = \begin{bmatrix} 7 \\ -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 7 \\ -2 \end{bmatrix} = 7 \begin{bmatrix} 1 \\ 0 \end{bmatrix} - 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 7 \\ -2 \end{bmatrix}$$

2) Find $\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}^{-1}$

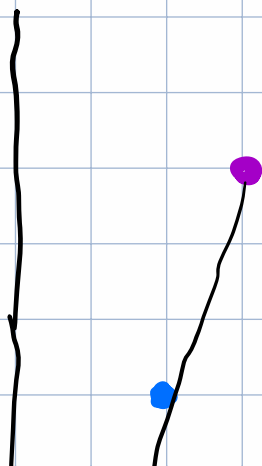
$$A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = 3I \rightarrow AA^{-1} = 3IA^{-1}$$
$$I = 3IA^{-1}$$
$$\frac{1}{3}I = A^{-1}$$
$$A^{-1} = \begin{bmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{3} \end{bmatrix}$$

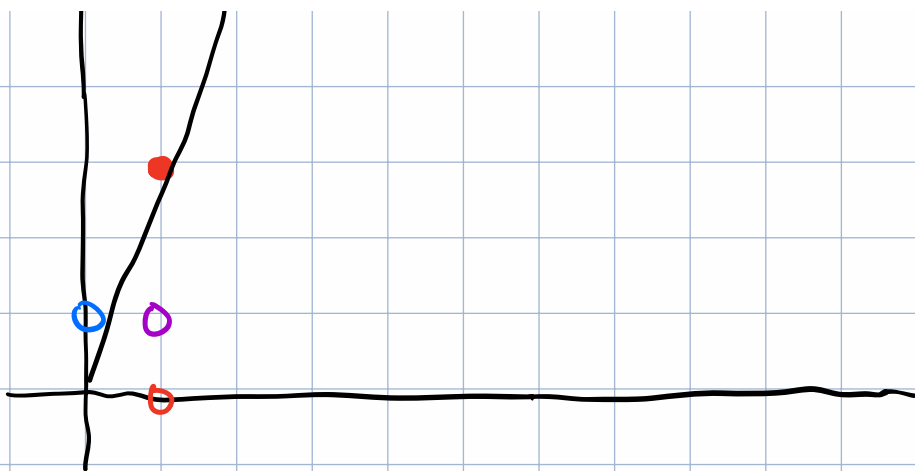
3) Without calculating anything else, is the following matrix invertible? Why?

$$\begin{matrix} c_1 & c_2 \\ \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \end{matrix}$$

$$c_2 = 2c_1$$

No, it is not invertible because all vectors are transformed onto a line, which has zero volume, so the determinant is zero. Also, since multiple vectors in our domain map to single values in the codomain, the transform is not bijective, and therefore the matrix is not invertible.





4) Suppose $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

which rotates vectors $\pi/2$ counter-clockwise.

What geometric transform should A^{-1} perform?

A^{-1} should rotate vectors $\pi/2$ clockwise.

$$A \approx i \rightarrow A^{-1} \approx -i$$

5) Why does a determinant of zero guarantee that no inverse exists?

See problem 3, essentially.

Any volume will transform to zero, so there is nothing we can multiply zero by to get

our original volume.

σ^-

σ^- U